Dept of Educational Psychology, Leadership, & Counseling

EPSY 5383

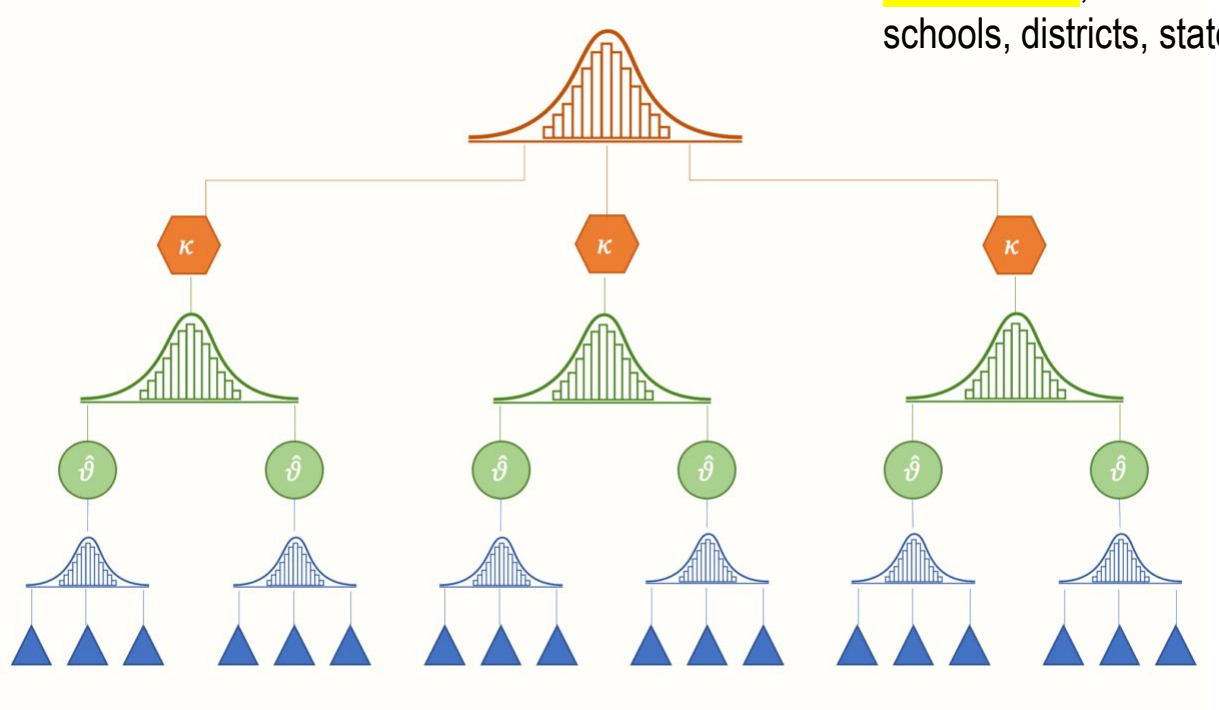
DATA ANALYSIS WITH STATISTICAL SOFTWARE

Module Overview

Hierarchical Data Structure

Hierarchical Data Structure

In education research, the data structure is often ***hierarchical***, with students *nested within* classrooms, schools, districts, states, and so on.



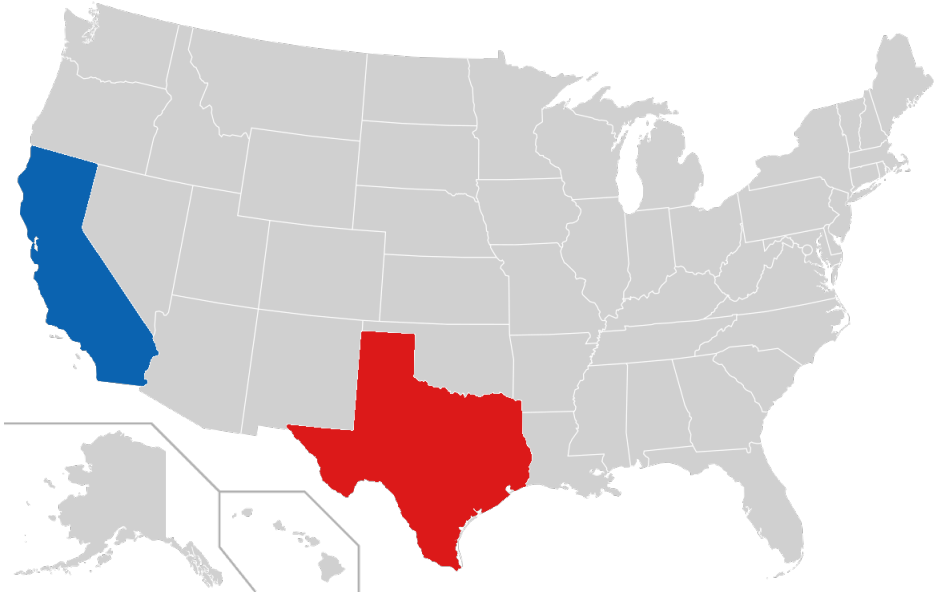
Hierarchical Data Structure

In education research, the data structure is often ***hierarchical***, with students *nested within* classrooms, schools, districts, states, and so on.

"Level 1"		**"Level 2"**
students		classrooms
states		U.S.
spouses	nested within	couples
cells		biopsy samples
repeated measures		individuals

- These are examples of *2-level* hierarchical data structure, in which lower-level (*level-1*) units are nested within higher-level (*level-2*) units.

Hierarchical Data Methods



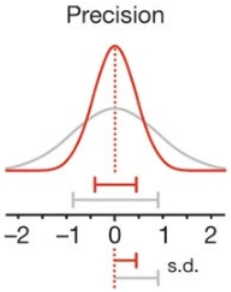
In multilevel data, cases *within clusters* are typically more similar than cases *between clusters*, which potentially violates the assumptions of…

- *Homoscedasticity*
- *Uncorrelated errors*

Hierarchical Data Methods

In multilevel data, sampling units no longer yield unique information. This leads to two primary problems:

1. *Underestimating SEs* of parameter estimates
   (increasing the chance of *falsely* rejecting the null hypothesis of no effect)



2. *Overestimating R^2*
   (overstating the current model's power to explain/predict the phenomenon)

Hierarchical Data Methods

Many different methods have been proposed to address the hierarchical nature of multilevel data. These methods fall into three rough categories depending on how they consider “nestedness”.

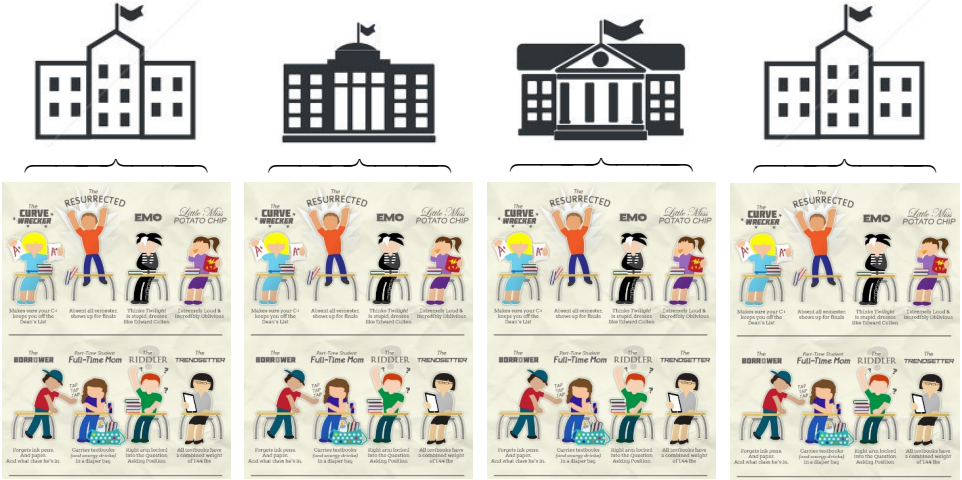
Disaggregation
Aggregation

} *ignore* nestedness

Ignore Nestedness

Level 2:
(schools)

Level 1:
(students)



Disaggregation

Level 1:
(students)

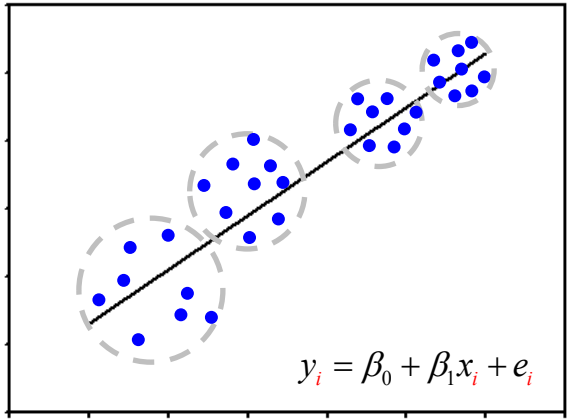
disaggregation to level 1
(ignore level 2)

Disaggregation

Disaggregation, also called ***pooled regression analysis***, involves analyzing data as if there is no clustering. Level-1 and level-2 predictors are included as if they are "independent".

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \beta_2 w_j + e_{ij} \quad \rightarrow \quad y_i = \beta_0 + \beta_1 x_i + \beta_2 w_i + e_i$$

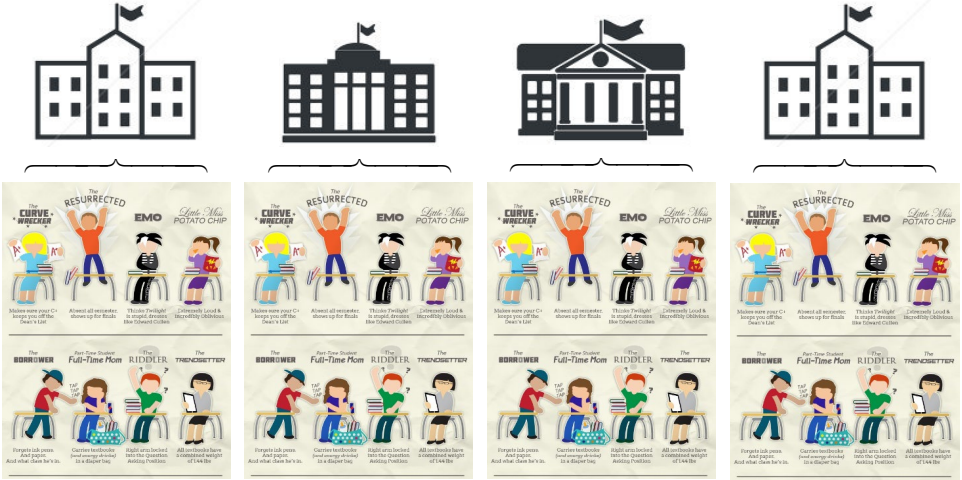
- Level-1 units are the unit of analysis.
- Disaggregation violates the assumption of *uncorrelated errors*, which can result in severe bias.
- This method also ignores *between-cluster variation*.



Ignore Nestedness

Level 2:
(schools)

Level 1:
(students)



Aggregation

Level 2:
(schools)

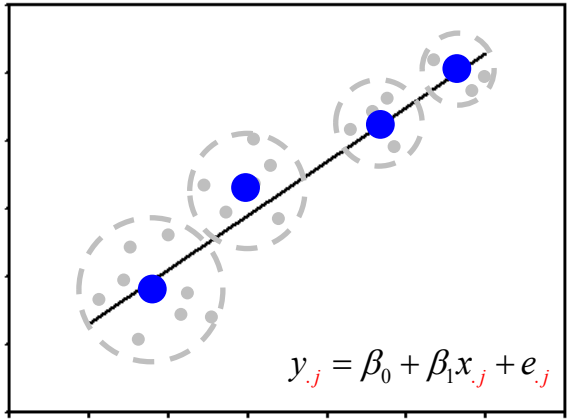
aggregation to level 2
(ignore level 1)

Aggregation

Aggregation involves "aggregating to level-2," or averaging data across members within each cluster.

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \beta_2 w_j + e_{ij} \quad \rightarrow \quad y_{.j} = \beta_0 + \beta_1 x_{.j} + \beta_2 w_j + e_{.j}$$

- Level-2 units are the unit of analysis; Sample size is the number of level-2 units.
- Clearly, aggregation can result in a massive *loss of information*.
- Interpretation is limited to level-2, which can be *misleading*.



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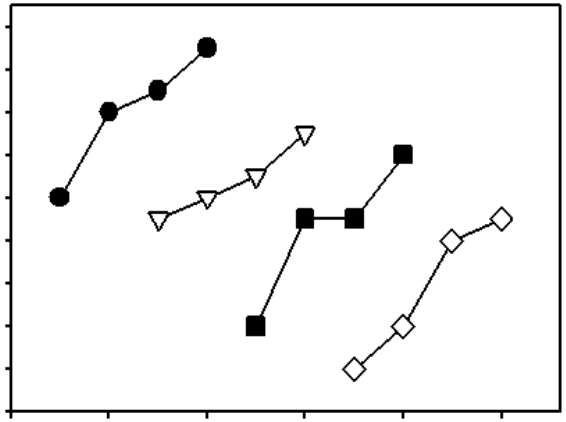
- "Anomalous effect"



Regression "within clusters" reveal *positive* trends, …

Aggregation

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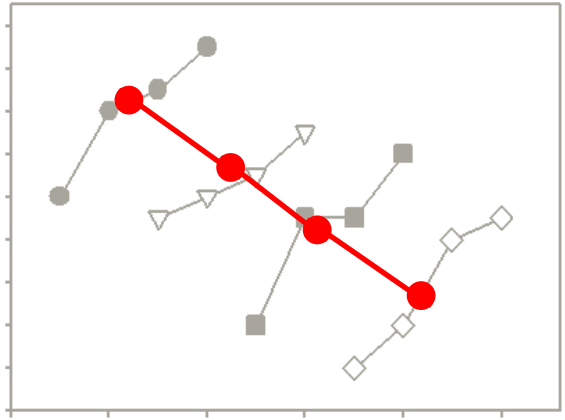
- "Anomalous effect"



Regression "within clusters" reveal *positive* trends, but if one averages "across clusters", a *negative* trend emerges.

Ignore Nestedness

The consequences of ignoring nestedness are (nearly) thoroughly discussed.

- For the case of regression analysis: Moerbeek (2004) and Opdenakker & Van Damme (2002)
- For the case of confirmatory factor analysis: Julian (2001) and Pornprasertmanit, Lee, & Preacher (2014)

Hierarchical Data Methods

Many different methods have been proposed to address the hierarchical nature of multilevel data. These methods fall into three rough categories depending on how they consider “nestedness”.

Disaggregation
Aggregation

} *ignore* nestedness

Adjusting the results
Fixed-effects regression
Separate regressions

} *control or correct for* nestedness

Hierarchical Data Methods

Many different methods have been proposed to address the hierarchical nature of multilevel data. These methods fall into three rough categories depending on how they consider “nestedness”.

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Adjusting the results
Fixed-effects regression
Separate regressions

} *control or correct for* nestedness

Multilevel modeling

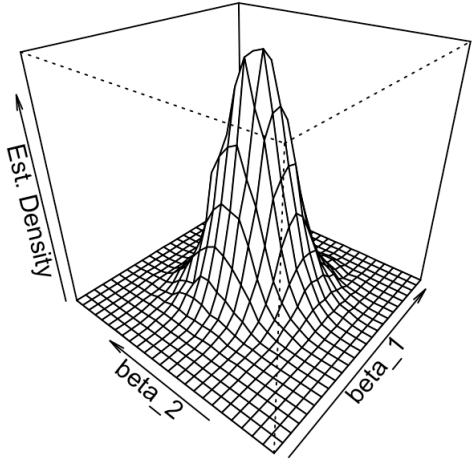
} *model* nestedness

Methods of dealing with multilevel data have moved gradually from *controlling for* nestedness (attempting to remove its influence) to *modeling* it as something worth studying.

Multilevel Modeling

Multilevel modeling (MLM) treats clusters as if they are *randomly* sampled from a large population of clusters, enhancing the *generalizability* of results.

- Cluster-level effects are not estimated separately for different clusters. Rather, the effects (i.e., regression coefficients) are assumed to have a particular "distribution" across clusters.



Multilevel Modeling

The advantages of MLM include…

- *Parsimony*. Rather than including separate cluster-level effects with dummy variables (i.e., fixed-effects regression), we simply estimate *mean and variance for an effect* across clusters.

- *Generalizability*. Because the effect is treated as ***random*** rather than fixed, we can *generalize* results to the population of clusters (e.g., schools, ISDs) rather than only to those sampled.

- *Flexibility*. By modeling nestedness, we can examine *both individual-level and cluster-level effects*.

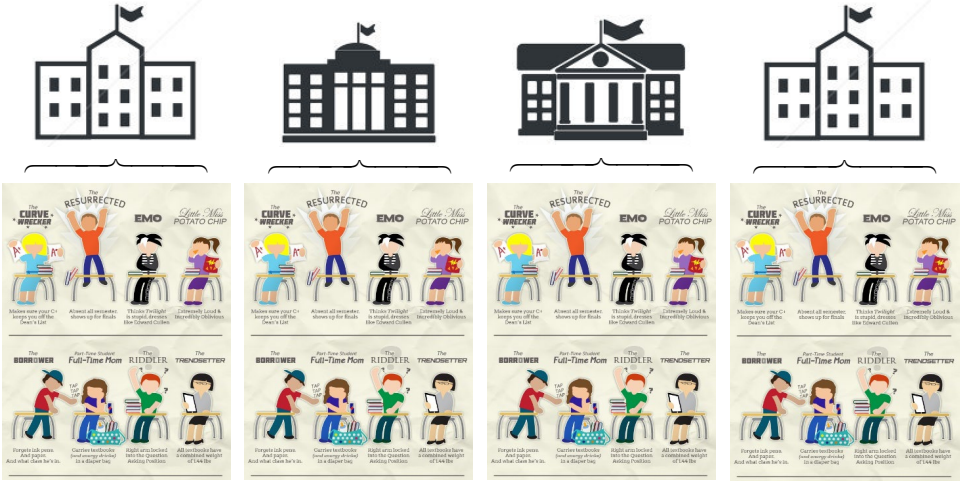
- *Appropriateness*. Multilevel models often conform more closely to theoretical predictions than do other approaches.

ICC & Notations

Null Model

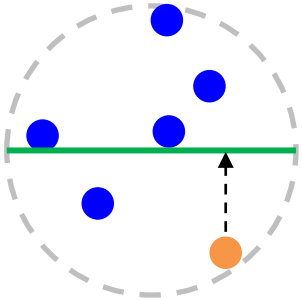
Level 2:
(schools)

Level 1:
(students)



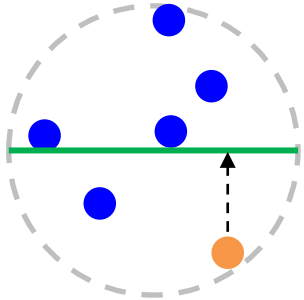
- You've decided to examine your data with multilevel modeling. Where do you start?
- It's always wise to begin with a "**null model**," another name of "**random intercept-only model**."

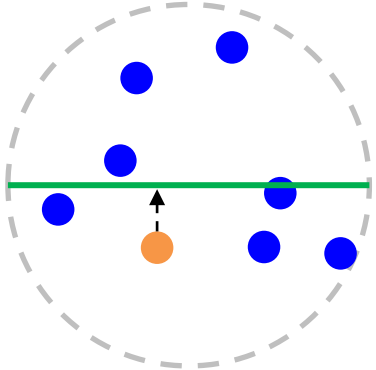
Null Model



My friend: "*My son got a math score that is 8 points below his school average.*"

Null Model

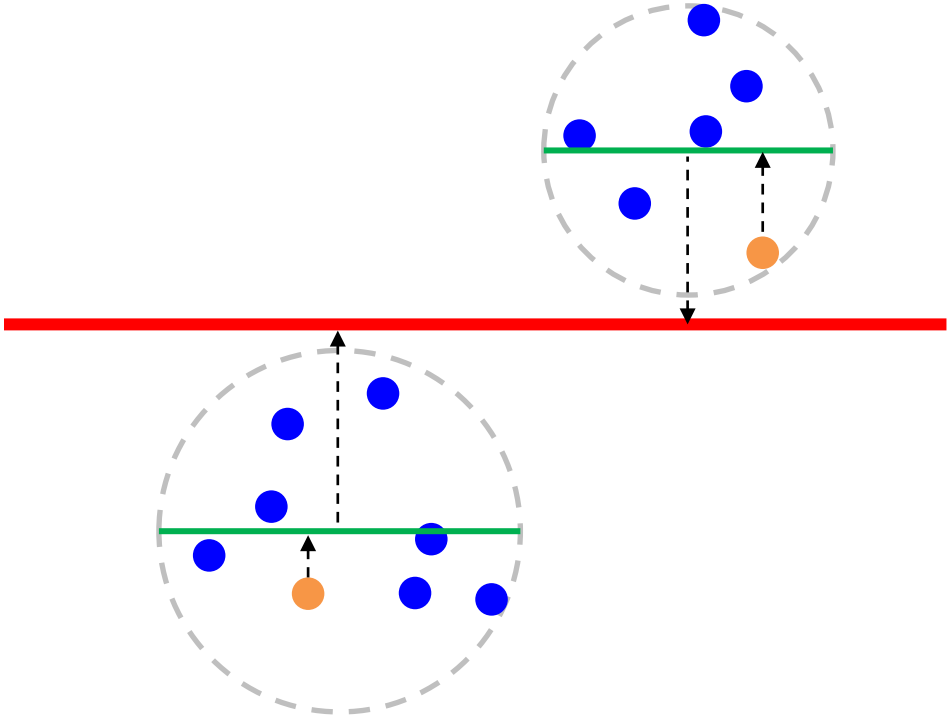




My friend: "*My son got a math score that is 8 points below his school average.*"

Me: "*Oh, that's too bad. My daughter's score is only 2 points below her school average (ha!). You should talk to your son.*"

Null Model

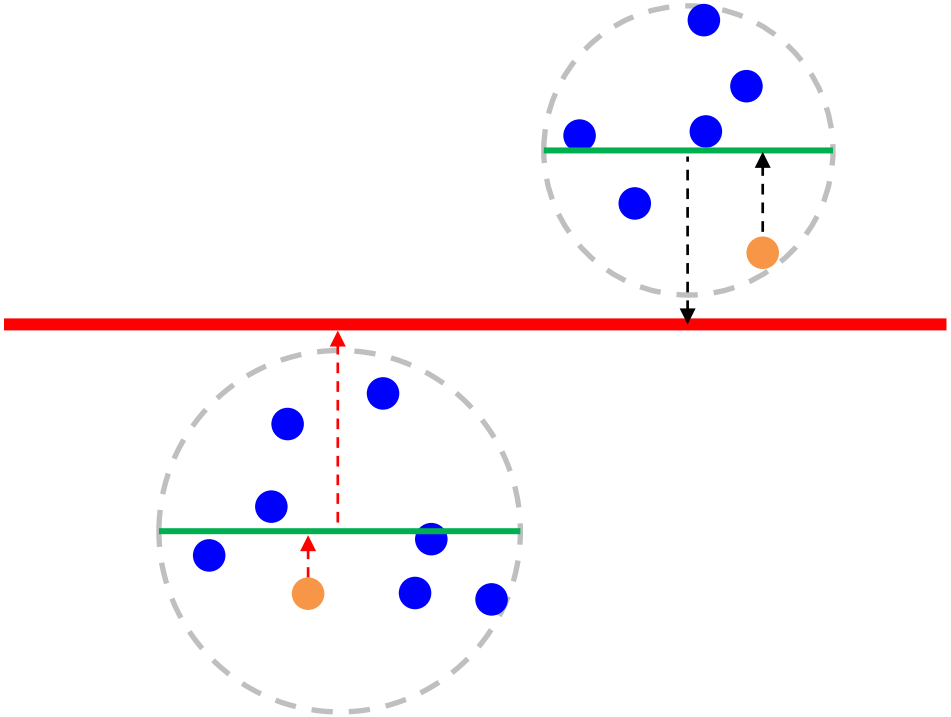


My friend: "*My son got a math score that is 8 points below his school average.*"

Me: "*Oh, that's too bad. My daughter's score is only 2 points below her school average (ha!). You should talk to your son.*"

My friend: "*Thanks for the advice. Yes, my son is struggling because his school is so competitive. His school average is 10 points higher than the state average. I think your daughter's school average is 15 points lower than the state average.*"

Null Model



My daughter's score = sum of …

the *state average*,

the *distance* from the state average to the school average, and

the *distance* from the school average to her score.

Null Model

Null model

Level-1 model: $y_{ij} = \beta_{0j} + e_{ij}$ | $e_{ij} \sim N\left(0, \sigma_e^2\right)$

- At level 1, for example, we express a student's score on y_{ij} as the sum of the student's *school mean* (β_{0j}; ***intercept***) and the student's *random deviation* (e_{ij}) from that mean.
  - σ_e^2 is the variance of level-1 residual, representing the *variability in student scores within the school*.

Null Model

Null model

Level-1 model: $y_{ij} = \beta_{0j} + e_{ij}$ | $e_{ij} \sim N\left(0, \sigma_e^2\right)$

Level-2 model: $\beta_{0j} = \gamma_{00} + u_{0j}$ | $u_{0j} \sim N\left(0, \tau_{00}\right)$

- At level 2, for example, we express the *school mean* (β_{0j}; ***intercept***) as the sum of the *state mean* (γ_{00}) and the school's *random deviation* (u_{0j}) from that mean.
  - γ_{00} is the *expected* value of the intercept, representing the *average score across all schools and all students* — i.e., ***grand mean*** in the state (population).
  - τ_{00} is the variance of level-2 residual, representing the *variability in school mean*.

Null Model

Null model

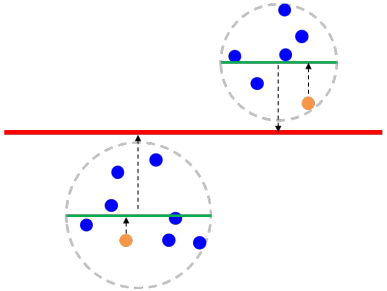
Level-1 model: $y_{ij} = \beta_{0j} + e_{ij}$ | $e_{ij} \sim N\left(0, \sigma_e^2\right)$

Level-2 model: $\beta_{0j} = \gamma_{00} + u_{0j}$ | $u_{0j} \sim N\left(0, \tau_{00}\right)$

Reduced-form equation: $y_{ij} = \gamma_{00} + \underbrace{u_{0j} + e_{ij}}$

"*fixed*" component

"*random*" components



- β_{0j} is a *"random" effect* because it has a *fixed* component (γ_{00}) and a *random* component (u_{0j}). In other words, for example, we want to generalize the results "beyond" the particular schools in the sample.

ICC

The *between*-cluster variance and the *within*-cluster variance sum to the *total* variance in y_{ij}:

$$y_{ij} = \beta_{0j} + e_{ij} \qquad e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$\beta_{0j} = \gamma_{00} + u_{0j} \qquad u_{0j} \sim N\left(0, \tau_{00}\right)$$

$$\hat{\tau}_{00} + \hat{\sigma}_e^2 = \hat{\sigma}_y^2$$

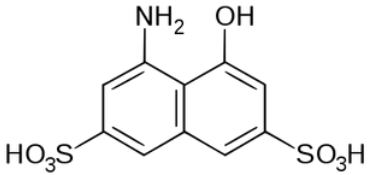
Intraclass correlation coefficient (ICC) is the "proportion" of *total variance* in in y_{ij} that is *attributable to the differences between clusters*.

$$\text{ICC} = \frac{\hat{\tau}_{00}}{\hat{\tau}_{00} + \hat{\sigma}_e^2} = \frac{\hat{\tau}_{00}}{\hat{\sigma}_y^2}$$

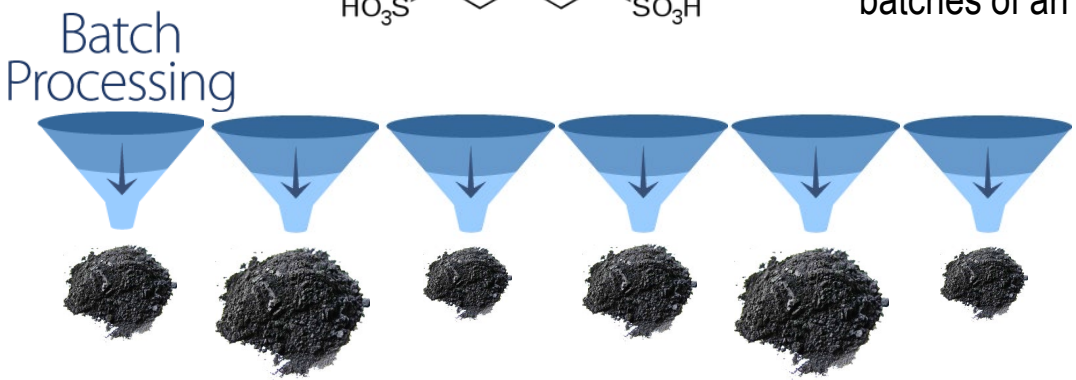
- ICC is often used to decide if MLM would be worthwhile. For example, with ICC below *.06*, the clustering of lower-level units can be ignored with little threat to validity. (Cheung & Au, 2005; Duncan, Alpert, & Duncan, 1998; Heck & Thomas, 2000; Muthén, 1989, 1991; Stapleton, 2006)

ICC

Example: dyestuff (Davies & Goldsmith, 1972)



- Dyestuff (Naphthalene Black 12B) from 6 different batches of an intermediate product (H-acid)



- The dyestuff production varies day by day. Is the observed variation mostly due to the differences between the batches or the differences within the batches?

ICC

Example: dyestuff

$$y_{ij} = \beta_{0j} + e_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$y_{ij} = \gamma_{00} + \underbrace{u_{0j} + e_{ij}}$$

"*fixed*" component

"*random*" components

R:

```
> null ← lmer(Yield ~ 1 + (1|Batch), data=Dyestuff, REML=FALSE)
```

- The terms before the parentheses describe the *fixed* components (parameters).
  - The "1" represents the ***intercept*** (γ_{00}).

ICC

Example: dyestuff

$$y_{ij} = \beta_{0j} + e_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$y_{ij} = \gamma_{00} + \underbrace{u_{0j} + e_{ij}}$$

"*fixed*" component

"*random*" components

R:

```
> null ← lmer(Yield ~ 1 + (1|Batch), data=Dyestuff, REML=FALSE)
```

- The terms in the parentheses describe the *random* components (parameters).
  - The "Batch" indicates that "batches" are clusters.
  - The "1" represents the ***intercept variance*** (τ_{00}).

ICC

Example: dyestuff

$$y_{ij} = \beta_{0j} + e_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$y_{ij} = \gamma_{00} + \underbrace{u_{0j} + e_{ij}}$$

"*fixed*" component

"*random*" components

R:

```
> null ← lmer(Yield ~ 1 + (1|Batch), data=Dyestuff, REML=FALSE)
```

- REML=FLASE changes the default choice of using REML to the *maximum likelihood (ML)* method for model parameter estimation.

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: Yield ~ 1 + (1 | Batch)
   Data: Dyestuff

     AIC      BIC   logLik deviance df.resid
   333.3    337.5   -163.7    327.3       27

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.4315 -0.7972  0.1480  0.7721  1.8037

Random effects:
 Groups   Name        Variance Std.Dev.
 Batch    (Intercept) 1388     37.26
 Residual             2451     49.51
Number of obs: 30, groups:  Batch, 6

Fixed effects:
            Estimate Std. Error t value
(Intercept)  1527.50      17.69   86.33
```

Level-2 variance: $\hat{\tau}_{00} = 1388(37.26)$

Level-1 variance: $\hat{\sigma}_e^2 = 2451(49.51)$

ICC: $$\frac{\hat{\tau}_{00}}{\hat{\tau}_{00} + \hat{\sigma}_e^2} = \frac{1388}{1388 + 2451} = 0.36$$

$$y_{ij} = \gamma_{00} + \underbrace{u_{0j} + e_{ij}}$$

"*fixed*" component

"*random*" components

- 36% of the total variance in the dyestuff production is due to the differences *between* batches.
- In other words, a greater proportion of the variation (64%) is due to the differences *within* batches.

NOTATION FOR VARIABLES

Variables

- Traditionally, ***y*** is used for *DV*s, ***x*** is used for *level-1 IV*s, and ***w*** is used for *level-2 IV*s.
  - y_{ij} — *DV* measured for a *level-1* unit *i* (e.g., student) within a *level-2* unit *j* (e.g., school).
  - x_{ij} — *IV* measured for a *level-1* unit *i* within a *level-2* unit *j*.
  - w_j — *IV* measured for a *level-2* unit *j*.

NOTATION FOR VARIABLES

Variables

- Traditionally, ***y*** is used for *DV*s, ***x*** is used for *level-1 IV*s, and ***w*** is used for *level-2 IV*s.
  - y_{ij} — e.g., *students'* STAAR test score, self-efficacy, anxiety, depression.
  - x_{ij} — e.g., *students'* age, gender, race, ethnicity, SES.
  - w_j — e.g., *schools'* female %, SES (free or reduced lunch %), size (# enrollment).

- DV is "always" measured *at the lowest level (i.e., level 1)*.

- *Number* subscripts are a part of the variable label but have nothing to do with the level at which it is measured. For example, x_{1ij} and x_{2ij} are simply two different level-1 IVs, and w_{1j} and w_{2j} are simply two different level-2 IVs.

NOTATION FOR COEFFICIENTS

Coefficients

$$y_{ij} = \beta_{0j} + \beta_{1j}x_{1ij} + \beta_{2j}x_{2ij} + e_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01}w_{1j} + \gamma_{02}w_{2j} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\beta_{2j} = \gamma_{20}$$

- There are as many subscripts attached to a coefficient symbol as there are levels of the data hierarchy.
  - 1 level $\longrightarrow$ 1 subscript;
  - 2 levels $\longrightarrow$ 2 subscripts;
  - 3 levels $\longrightarrow$ 3 subscripts, and so on.

NOTATION FOR COEFFICIENTS

Coefficients

$$y_{ij} = \beta_{0j} + \beta_{1j}x_{1ij} + \beta_{2j}x_{2ij} + e_{ij}$$

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$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\beta_{2j} = \gamma_{20}$$

- <u>*Number*</u> subscripts indicate the <u>*position*</u> of the coefficients in the level-1, level-2, and level-3 equations, starting with "0."
  - The "0" in $\boldsymbol{\beta_{0j}}$ indicates that this regression coefficient is the ***intercept*** in the level-1 equation.
  - The "1" in $\boldsymbol{\beta_{1j}}$ indicates that this coefficient is the "first" ***slope*** in the level-1 equation.
  - The "2" in $\boldsymbol{\beta_{2j}}$ indicate that this coefficient is the "second" ***slope*** in the level-1 equation.

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Coefficients

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$$\beta_{2j} = \gamma_{20}$$

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- Note that all the subscripts begin with "0" in the level-2 equation for the ***intercept*** $\boldsymbol{\beta_{0j}}$.

NOTATION FOR COEFFICIENTS

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- Note that all the subscripts begin with "0" in the level-2 equation for the ***intercept*** $\boldsymbol{\beta_{0j}}$.
- The "1" in $\boldsymbol{\gamma_{01}}$ indicates that this regression coefficient is the "first" ***slope*** in the level-2 equation for the *intercept* $\boldsymbol{\beta_{0j}}$.
- The "2" in $\boldsymbol{\gamma_{02}}$ indicates that this coefficient is the "second" ***slope*** in the level-2 equation for the *intercept* $\boldsymbol{\beta_{0j}}$.

NOTATION FOR COEFFICIENTS

Coefficients

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$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\beta_{2j} = \gamma_{20}$$

- The "1" in $\boldsymbol{\beta_{1j}}$ indicates that this regression coefficient is the "first" ***slope*** in the level-1 equation.
- Note that all the subscripts begin with "1" in the level-2 equation for the ***slope*** $\boldsymbol{\beta_{1j}}$.
- The "0" in γ_{10} indicates that this regression coefficient is the ***intercept*** in the level-2 equation for the *slope* $\boldsymbol{\beta_{1j}}$.

NOTATION FOR COEFFICIENTS

Coefficients

- For *random* ***variance*** components, the subscripts "clone" the integer value of the first subscript for the corresponding fixed component (e.g., 00, 11, 22, etc.).

- ***Covariances*** are indicated by a combination of two different subscript values (e.g., 10, 20, 21, etc.).

$$y_{ij} = \beta_{0j} + \beta_{1j} x_{1ij} + \beta_{2j} x_{2ij} + u_{0j} + u_{1j} x_{1ij} + e_{ij}$$

$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{00} & \\ \tau_{10} & \tau_{11} \end{bmatrix} \right)$$

- τ_{11} represents the ***variance*** of the "first" *slope* in the level-1 equation – e.g., how much the slope value varies between students.

- τ_{10} represents the ***covariance*** between β_{0j} and β_{1j}.

NOTATION FOR COEFFICIENTS

Coefficients

- A complicated example: 2-level model with three level-1 IVs, a random intercept and all random slopes, and five level-2 IVs:

$$y_{ij} = \beta_{0j} + \beta_{1j}x_{1ij} + \beta_{2j}x_{2ij} + \beta_{3j}x_{3ij} + e_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01}w_{1j} + \gamma_{02}w_{2j} + \gamma_{03}w_{3j} + \gamma_{04}w_{4j} + \gamma_{05}w_{5j} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}w_{1j} + \gamma_{12}w_{2j} + \gamma_{13}w_{3j} + \gamma_{14}w_{4j} + \gamma_{15}w_{5j} + u_{1j}$$

$$\beta_{2j} = \gamma_{20} + \gamma_{21}w_{1j} + \gamma_{22}w_{2j} + \gamma_{23}w_{3j} + \gamma_{24}w_{4j} + \gamma_{25}w_{5j} + u_{2j}$$

$$\beta_{3j} = \gamma_{30} + \gamma_{31}w_{1j} + \gamma_{32}w_{2j} + \gamma_{33}w_{3j} + \gamma_{34}w_{4j} + \gamma_{35}w_{5j} + u_{3j}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$\begin{bmatrix} u_{0j} \\ u_{1j} \\ u_{2j} \\ u_{3j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{00} & & & \\ \tau_{10} & \tau_{11} & & \\ \tau_{20} & \tau_{21} & \tau_{22} & \\ \tau_{30} & \tau_{31} & \tau_{32} & \tau_{33} \end{bmatrix}\right)$$

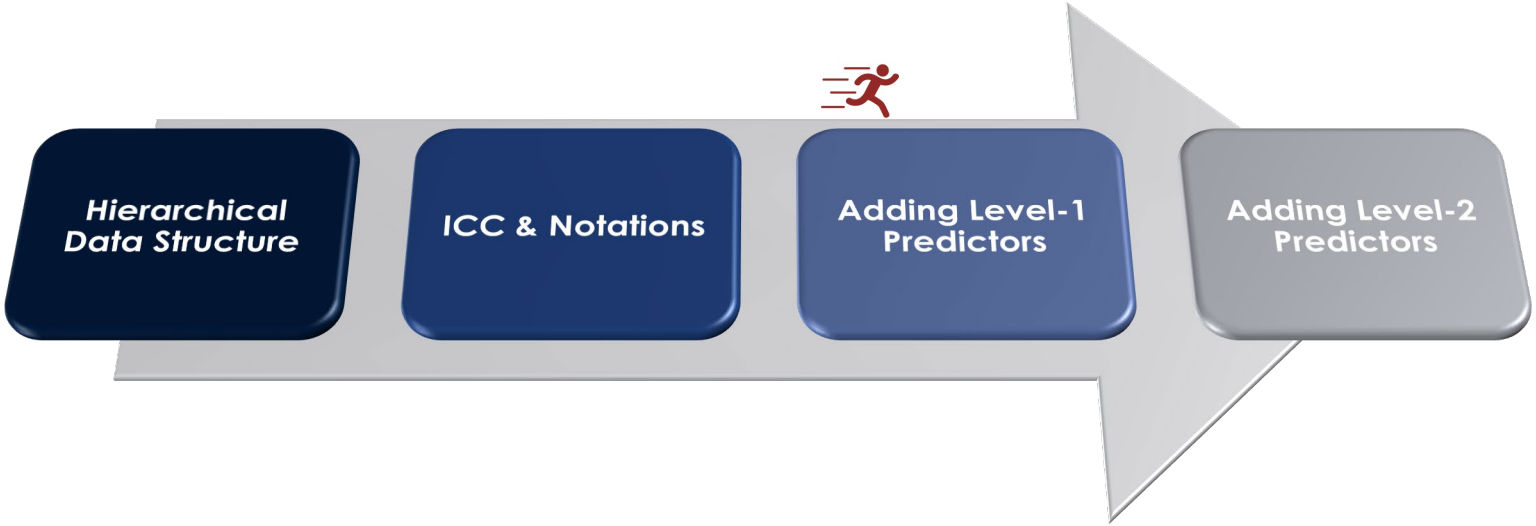
NOTATION FOR COEFFICIENTS

Coefficients

- The same model in reduced form:

$$
\begin{aligned}
y_{ij} = {} & \gamma_{00} + \gamma_{10} x_{1ij} + \gamma_{20} x_{2ij} + \gamma_{30} x_{3ij} + \\
& \gamma_{01} w_{1j} + \gamma_{11} x_{1ij} w_{1j} + \gamma_{21} x_{2ij} w_{1j} + \gamma_{31} x_{3ij} w_{1j} + \\
& \gamma_{02} w_{2j} + \gamma_{12} x_{1ij} w_{2j} + \gamma_{22} x_{2ij} w_{2j} + \gamma_{32} x_{3ij} w_{2j} + \\
& \gamma_{03} w_{3j} + \gamma_{13} x_{1ij} w_{3j} + \gamma_{23} x_{2ij} w_{3j} + \gamma_{33} x_{3ij} w_{3j} + \\
& \gamma_{04} w_{4j} + \gamma_{14} x_{1ij} w_{4j} + \gamma_{24} x_{2ij} w_{4j} + \gamma_{34} x_{3ij} w_{4j} + \\
& \gamma_{05} w_{5j} + \gamma_{15} x_{1ij} w_{5j} + \gamma_{25} x_{2ij} w_{5j} + \gamma_{35} x_{3ij} w_{5j} + \\
& u_{0j} + u_{1j} x_{1ij} + u_{2j} x_{2ij} + u_{3j} x_{3ij} + e_{ij}
\end{aligned}
$$

Adding Level-1 Predictors



Null Model

Null model

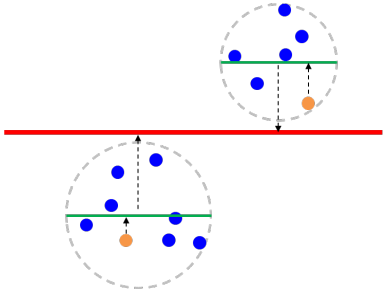
Level-1 model: $y_{ij} = \beta_{0j} + e_{ij}$ | $e_{ij} \sim N\left(0, \sigma_e^2\right)$

Level-2 model: $\beta_{0j} = \gamma_{00} + u_{0j}$ | $u_{0j} \sim N\left(0, \tau_{00}\right)$

Reduced-form equation: $y_{ij} = \gamma_{00} + \underbrace{u_{0j} + e_{ij}}$

"*fixed*" component

"*random*" components



Fixed vs. Random

We can expand the null model to include a *fixed* *level-1* predictor.

$$y_{ij} = \beta_{0j} + \beta_{1j} x_{ij} + e_{ij}$$
$$\beta_{0j} = \gamma_{00} + u_{0j}$$
$$\beta_{1j} = \gamma_{10}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$
$$u_{0j} \sim N\left(0, \tau_{00}\right)$$

Reduced-form equation: $y_{ij} = \gamma_{00} + \gamma_{10} x_{ij} + u_{0j} + e_{ij}$

- For example, the effect of ***x*** is constrained to be the *same across the schools* — i.e., *fixed* ***slope***, $\tau_{11} = 0$.

Fixed vs. Random

We can expand the null model to include a *fixed* *level-1* predictor.

$$y_{ij} = \beta_{0j} + \beta_{1j} x_{ij} + e_{ij}$$
$$\beta_{0j} = \gamma_{00} + u_{0j}$$
$$\beta_{1j} = \gamma_{10}$$

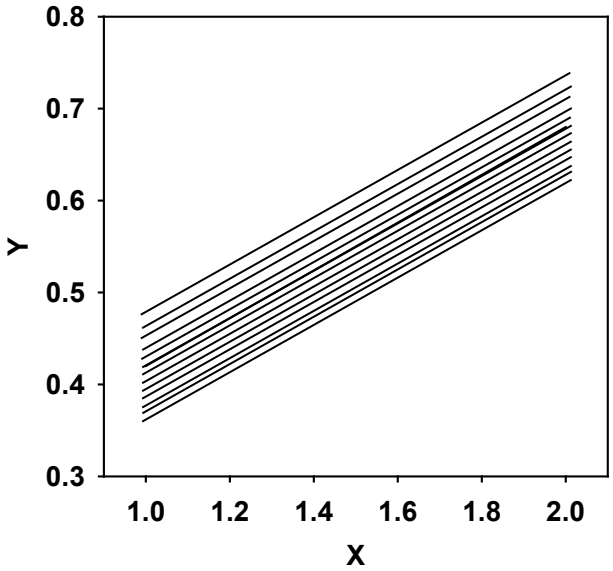
$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$
$$u_{0j} \sim N\left(0, \tau_{00}\right)$$

Reduced-form equation: $y_{ij} = \gamma_{00} + \gamma_{10} x_{ij} + u_{0j} + e_{ij}$

- For example, the effect of ***x*** is constrained to be the *same across the schools* — i.e., *fixed* ***slope*** (τ_{11} = 0).
- School means, on the other hand, are allowed to *vary across the schools* — i.e., *random* ***intercepts*** (τ_{00} > 0).

Fixed vs. Random

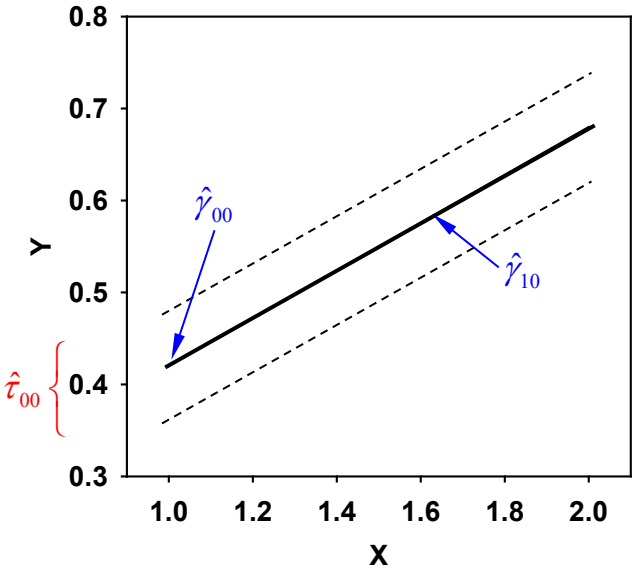
random intercept, fixed slope



Fixed vs. Random

$$y_{ij} = \gamma_{00} + \gamma_{10} x_{ij} + u_{0j} + e_{ij}$$

random intercept, fixed slope



Fixed vs. Random

We can further expand the model by including a *random* variance component for the *level-1* predictor:

$$y_{ij} = \beta_{0j} + \beta_{1j} x_{ij} + e_{ij}$$
$$\beta_{0j} = \gamma_{00} + u_{0j}$$
$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$
$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{00} & \\ \tau_{10} & \tau_{11} \end{bmatrix}\right)$$

Reduced-form equation:

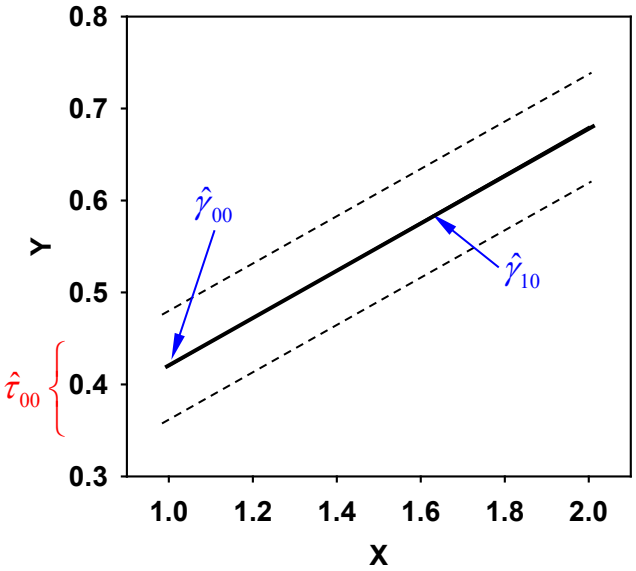
$$y_{ij} = \gamma_{00} + \gamma_{10} x_{ij} + u_{0j} + u_{1j} x_{ij} + e_{ij}$$

- For example, the effect of ***x*** is allowed to *vary across schools* — i.e., *random* ***slope*** ($\tau_{11} > 0$).

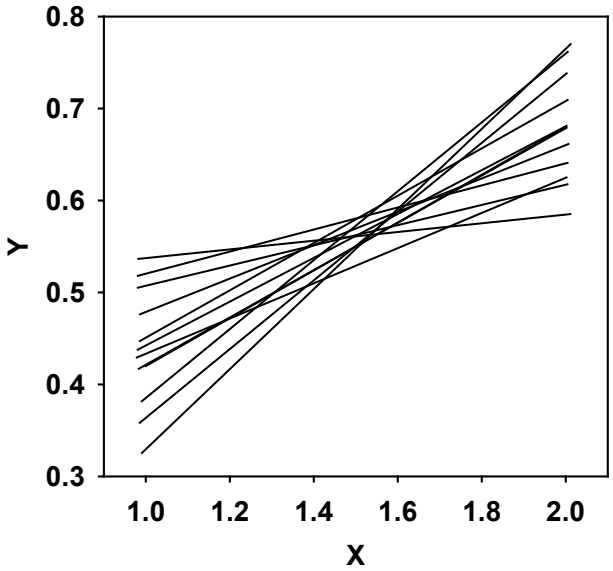
Fixed vs. Random

$$y_{ij} = \gamma_{00} + \gamma_{10} x_{ij} + u_{0j} + e_{ij}$$

random intercept, fixed slope



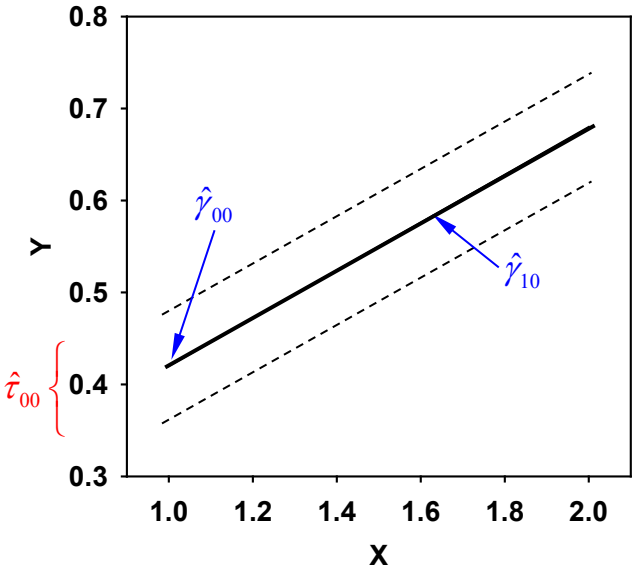
random intercept, random slope



Fixed vs. Random

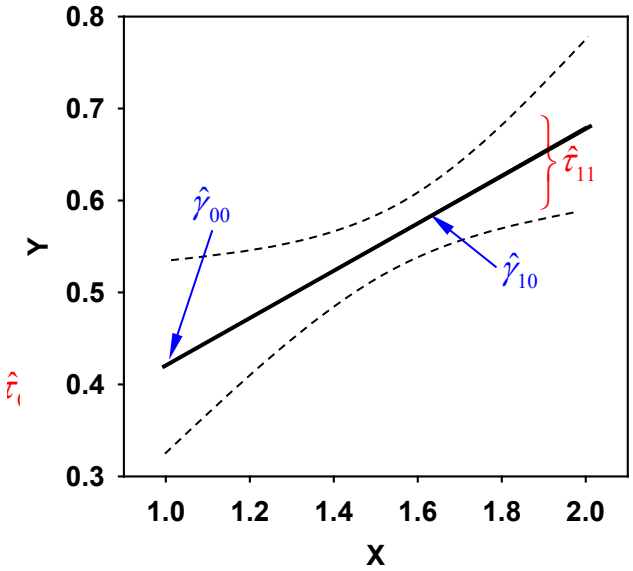
$$y_{ij} = \gamma_{00} + \gamma_{10} x_{ij} + u_{0j} + e_{ij}$$

random intercept, fixed slope



$$y_{ij} = \gamma_{00} + \gamma_{10} x_{ij} + u_{0j} + u_{1j} x_{ij} + e_{ij}$$

random intercept, random slope



Adding Level-1 Predictors

Example: popular2 (Hox, 2010, Appendix A)

- 2,000 students in 100 classrooms. Average class size is 20.
  - Popularity (*popular*: range = 0–10): average of popularity ratings by classmates.
  - Extraversion (*extrav*: range = 0–10)

Adding Level-1 Predictors

Null model

$$popular_{ij} = \gamma_{00} + u_{0j} + e_{ij} \quad \Bigg| \quad \begin{aligned} e_{ij} &\sim N\left(0, \sigma_e^2\right) \\ u_{0j} &\sim N\left(0, \tau_{00}\right) \end{aligned}$$

R:

```
> null ← lmer(popular ~ 1 + (1|class), data=dat)
> summary(null)
> confint(null, method="profile", level=0.95)
```

- The terms before the parentheses describe the *fixed* components (parameters).
  - The "1" represents the ***intercept*** (γ_{00}).

Adding Level-1 Predictors

Null model

$$popular_{ij} = \gamma_{00} + u_{0j} + e_{ij}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$u_{0j} \sim N\left(0, \tau_{00}\right)$$

R:

```
> null ← lmer(popular ~ 1 + (1|class), data=dat)
> summary(null)
> confint(null, method="profile", level=0.95)
```

- The terms in the parentheses describe the *random* components (parameters).
  - The "class" indicates that "classrooms" are clusters.
  - The "1" represents the ***intercept variance*** (τ_{00}).

Adding Level-1 Predictors

Null model

$$popular_{ij} = \gamma_{00} + u_{0j} + e_{ij} \quad \Bigg| \quad \begin{aligned} e_{ij} &\sim N\left(0, \sigma_e^2\right) \\ u_{0j} &\sim N\left(0, \tau_{00}\right) \end{aligned}$$

R:

```
> null ← lmer(popular ~ 1 + (1|class), data=dat)
> summary(null)
> confint(null, method="profile", level=0.95)
```

- confint() computes confidence intervals using:
  - “profile”: Likelihood function
  - “Wald”: Fixed parameters only
  - “boot”: Parametric bootstrapping

Predictors

```
> summary(null)
Linear mixed model fit by REML ['lmerMod']
Formula: popular ~ 1 + (1 | class)
   Data: dat

REML criterion at convergence: 6330.5

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.5655 -0.6975  0.0020  0.6758  3.3175

Random effects:
 Groups   Name        Variance Std.Dev.
 class    (Intercept) 0.7021   0.8379
 Residual             1.2218   1.1053
Number of obs: 2000, groups:  class, 100

Fixed effects:
            Estimate Std. Error t value
(Intercept)  5.07786    0.08739    58.1
> # Confidence intervals of model parameters
> confint(null, method = "profile", level = 0.95)
Computing profile confidence intervals ...
                2.5 %    97.5 %
.sig01      0.719963 0.9744636
.sigma      1.071113 1.1414463
(Intercept) 4.905780 5.2499439
```

$$popular_{ij} = \gamma_{00} + u_{0j} + e_{ij}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$u_{0j} \sim N\left(0, \tau_{00}\right)$$

$$\hat{\gamma}_{00} = 5.08\ (0.09),\ p < .001$$

- The average popularity score across all classes and all students

$$\hat{\sigma}_e^2 = 1.22\ [1.07;\ 1.14]$$

$$\hat{\tau}_{00} = 0.70\ [0.72;\ 0.97]$$

- Significant ***intercept variance*** — i.e., the popularity score significantly varied across the classes.

$$\text{ICC} = \frac{\hat{\tau}_{00}}{\hat{\tau}_{00} + \hat{\sigma}_e^2} = \frac{0.70}{0.70 + 1.22} = 0.36$$

- 36% of the total variance in the popularity score occurred at the class level, which is substantial.

Adding Level-1 Predictors

Adding 'extrav': Random intercept, <u>fixed</u> slope

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + u_{0j} + e_{ij}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$u_{0j} \sim N\left(0, \tau_{00}\right)$$

<u>R</u>:

```
> l1_fs ← lmer(popular ~ 1 + extrav + (1|class), data=dat)
> summary(l1_fs)
> confint(l1_fs, method="profile", level=0.95)
```

- The terms before the parentheses describe the *fixed* components (parameters).
  - The "extrav" represents the ***slope*** (γ_{10}) for the extraversion score.

```
> summary(l1_fs)
Linear mixed model fit by REML ['lmerMod']
Formula: popular ~ 1 + extrav + (1 | class)
   Data: dat

REML criterion at convergence: 5832.6

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.0644 -0.7267  0.0165  0.7088  3.3587

Random effects:
 Groups   Name        Variance Std.Dev.
 class    (Intercept) 0.8406   0.9168
 Residual             0.9304   0.9646
Number of obs: 2000, groups:  class, 100

Fixed effects:
            Estimate Std. Error t value
(Intercept)  2.54214    0.14113   18.01
extrav       0.48631    0.02015   24.13

Correlation of Fixed Effects:
       (Intr)
extrav -0.745
> ### Confidence intervals of model parameters
> confint(l1_fs, method="profile", level=0.95)
Computing profile confidence intervals ...
                 2.5 %    97.5 %
.sig01       0.7913536 1.0624251
.sigma       0.9344460 0.9958162
(Intercept) 2.2647455 2.8187560
extrav       0.4465543 0.5258169
```

Predictors

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + u_{0j} + e_{ij}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$u_{0j} \sim N\left(0, \tau_{00}\right)$$

$$\hat{\gamma}_{00} = 2.54\ (0.14),\ p < .001$$

- The average popularity score for students whose extraversion score = 0

$$\hat{\gamma}_{10} = 0.49\ (0.02),\ p < .001$$

- The popularity score is “expected” to increase by 0.49 points with each 1-point increase in the extraversion score.

```
> summary(l1_fs)
Linear mixed model fit by REML ['lmerMod']
Formula: popular ~ 1 + extrav + (1 | class)
   Data: dat

REML criterion at convergence: 5832.6

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.0644 -0.7267  0.0165  0.7088  3.3587

Random effects:
 Groups   Name        Variance Std.Dev.
 class    (Intercept) 0.8406   0.9168
 Residual             0.9304   0.9646
Number of obs: 2000, groups:  class, 100

Fixed effects:
            Estimate Std. Error t value
(Intercept)  2.54214    0.14113   18.01
extrav       0.48631    0.02015   24.13

Correlation of Fixed Effects:
       (Intr)
extrav -0.745
> ### Confidence intervals of model parameters
> confint(l1_fs, method="profile", level=0.95)
Computing profile confidence intervals ...
                 2.5 %    97.5 %
.sig01       0.7913536 1.0624251
.sigma       0.9344460 0.9958162
(Intercept) 2.2647455 2.8187560
extrav       0.4465543 0.5258169
```

Predictors

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + u_{0j} + e_{ij}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$u_{0j} \sim N\left(0, \tau_{00}\right)$$

$$\hat{\sigma}_e^2 = 0.93 \ [0.93;\ 1.00]$$

$$\hat{\tau}_{00} = 0.84 \ [0.79;\ 1.06]$$

$$\text{ICC}_R = \frac{\hat{\tau}_{00}}{\hat{\tau}_{00} + \hat{\sigma}_e^2} = \frac{0.84}{0.84 + 0.93} = 0.47$$

- 47% of the “residual” variance in the popularity scores was explained at the class level, which is still substantial.

Adding Level-1 Predictors

Adding 'extrav': Random intercept, <u>random</u> slope

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + u_{0j} + u_{1j} extrav_{ij} + e_{ij}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{00} & \\ \tau_{10} & \tau_{11} \end{bmatrix}\right)$$

<u>R</u>:

```
> l1_rs ← lmer(popular ~ 1 + extrav + (1+ extrav|class),
data=dat)
> summary(l1_rs)
> confint(l1_rs, method="profile", level=0.95)
```

- The terms in the parentheses describe the *random* components (parameters).
  - The "extrav" represents the ***slope variance*** (τ_{11}).
  - The model also includes the ***intercept-slope covariance*** (τ_{11}).

.1 Predictors

```
Warning: Model failed to converge with max|grad| = 0.
> summary(l1_rs)
Linear mixed model fit by REML ['lmerMod']
Formula: popular ~ 1 + extrav + (1 + extrav | class)
   Data: dat

REML criterion at convergence: 5779.4

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.1961 -0.7291  0.0145  0.6817  3.2216

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 class    (Intercept) 2.99680  1.7311
          extrav      0.02595  0.1611   -0.97
 Residual             0.89495  0.9460
Number of obs: 2000, groups:  class, 100

Fixed effects:
            Estimate Std. Error t value
(Intercept)  2.46106    0.20309   12.12
extrav       0.49286    0.02545   19.36

Correlation of Fixed Effects:
       (Intr)
extrav -0.917
optimizer (nloptwrap) convergence code: 0 (OK)
Model failed to converge with max|grad| = 0.00663866

> ### Confidence intervals of model parameters
> confint(l1_rs, method="profile", level=0.95)
Computing profile confidence intervals ...
                 2.5 %     97.5 %
.sig01       1.4156977  2.0829902
.sig02      -1.0000000 -0.8989131
.sig03       0.1056739  0.2170604
.sigma       0.9159564  0.9778293
(Intercept)  2.0583720  2.8594566
extrav       0.4422616  0.5429940
```

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + u_{0j} + u_{1j} extrav_{ij} + e_{ij}$$

$$\hat{\tau}_{00} = 3.00 \ [1.42;\ 2.08]$$

- Significant ***intercept variance*** — i.e., the popularity score significantly varied across the classes.

$$\hat{\tau}_{11} = 0.03 \ [0.11;\ 0.22]$$

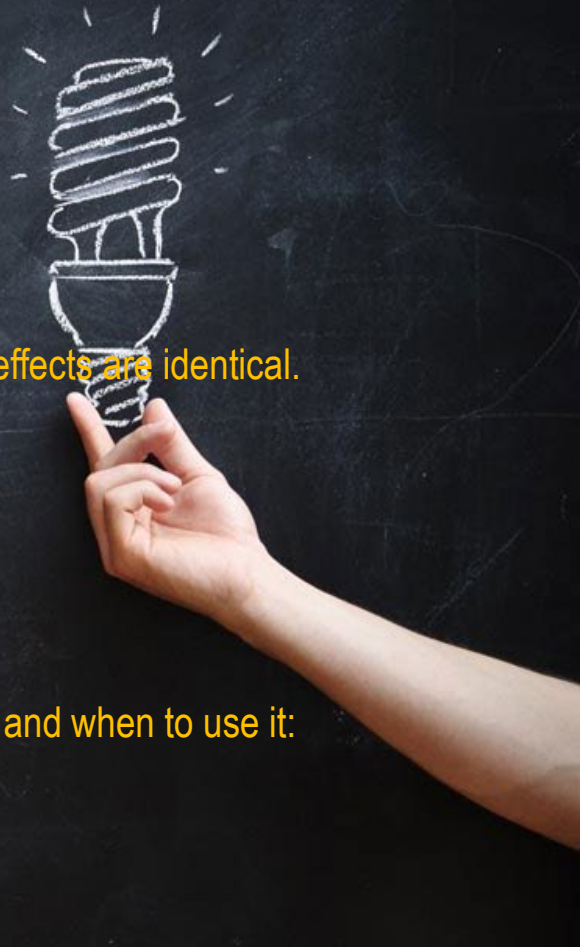
- Significant ***slope variance*** — i.e., the slope for (effect of) the extraversion score significantly varied across the classes.

$$\hat{\tau}_{10} = -0.97 \ [-1.00;\ -0.90]$$

- Significant ***intercept-slope covariance*** — ?

The default parameter estimation criterion for multilevel models is ***restricted (or "residual") maximum likelihood (REML)***.

- ML estimates of ***variance components*** are known to be *biased*.
- Although REML estimates are not guaranteed to be unbiased, they are usually *less biased* than ML estimates.
- For a balanced, one-way classification, the REML and ML estimates of the fixed-effects are identical.



You may want to check this out, pretty good explanation of basics of what REML is and when to use it: http://users.stat.umn.edu/~gary/classes/5303/handouts/REML.pdf.

Likelihood-Ratio Test

Let's consider two models, $M_0 < M_1$. **Likelihood-ratio test (LRT)** compares two "nested" models by comparing their ***deviances***:

$$-2\left(L_0 - L_1\right) = -2\left(L_0 - L_S\right) - 2\left(L_1 - L_S\right)$$

- This test statistic is assumed to follow a chi-square (χ^2) distribution, with *df* = the number of additional parameters that are in M_1 but not in M_0.
  - A *large* statistics value (therefore, a *small* *p*-value) suggests that the more complex model M_1 fits <u>*significantly better*</u> than the simpler model $M_0 \rightarrow$ retain M_1 for the sake of *model fit*.
  - A *small* statistics value (therefore, a *large* *p*-value) suggests that the simpler model M_0 fits <u>*as good as*</u> the more complex model $M_1 \rightarrow$ retain M_0 for the sake of *model parsimony*.

```
> anova(null, l1_fs)
refitting model(s) with ML (instead of REML)
Data: dat
Models:
null: popular ~ 1 + (1 | class)
l1_fs: popular ~ 1 + extrav + (1 | class)
      npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
null     3 6333.5 6350.3 -3163.7   6327.5
l1_fs    4 5831.8 5854.2 -2911.9   5823.8 503.69  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> anova(l1_fs, l1_rs)
refitting model(s) with ML (instead of REML)
Data: dat
Models:
l1_fs: popular ~ 1 + extrav + (1 | class)
l1_rs: popular ~ 1 + extrav + (1 + extrav | class)
      npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
l1_fs    4 5831.8 5854.2 -2911.9   5823.8
l1_rs    6 5782.7 5816.3 -2885.3   5770.7 53.089  2  2.964e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

$$popular_{ij} = \gamma_{00} + u_{0j} + e_{ij}$$

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + u_{0j} + e_{ij}$$

R:

```
> anova(null, l1_fa)
```

- LRT $\chi^2(1) = 503.69$, $p < .001$ suggests that the “random-intercept-fixed-slope” model fits significantly better than the “random-intercept-only” null model.
- This provides additional support for the effect of extraversion on popularity.

```
> anova(null, l1_fs)
refitting model(s) with ML (instead of REML)
Data: dat
Models:
null: popular ~ 1 + (1 | class)
l1_fs: popular ~ 1 + extrav + (1 | class)
      npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
null     3 6333.5 6350.3 -3163.7   6327.5
l1_fs    4 5831.8 5854.2 -2911.9   5823.8 503.69  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> anova(l1_fs, l1_rs)
refitting model(s) with ML (instead of REML)
Data: dat
Models:
l1_fs: popular ~ 1 + extrav + (1 | class)
l1_rs: popular ~ 1 + extrav + (1 + extrav | class)
      npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
l1_fs    4 5831.8 5854.2 -2911.9   5823.8
l1_rs    6 5782.7 5816.3 -2885.3   5770.7 53.089  2  2.964e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + u_{0j} + e_{ij}$$

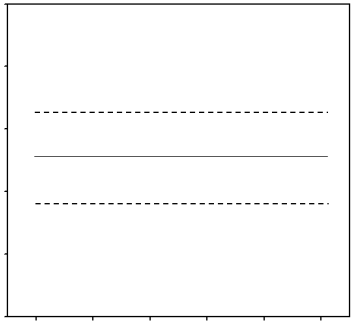
$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + u_{0j} + u_{1j} extrav_{ij} + e_{ij}$$

R:

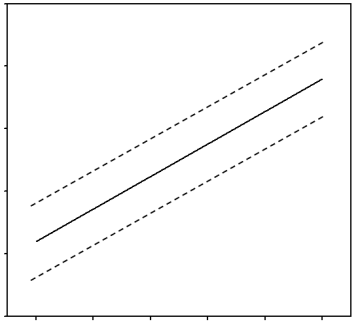
```
> anova(l1_fa, l1_rs)
```

- LRT $\chi^2(2) = 53.09$, $p < .001$ suggests that the “random-intercept-random-slope” model fits significantly better than the “random-intercept-fixed-slope” model.
- This provides additional support for the fact that the effect of extraversion varied across the classes.

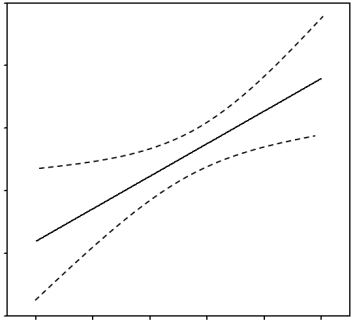
Likelihood-Ratio Test



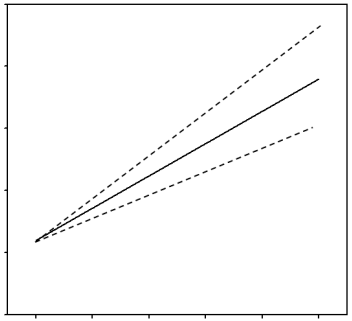
M_0: *random* intercept only



M_{1a} : *random* intercept, *fixed* slope



M_2: *random* intercept, *random* slope



M_{1b}: *fixed* intercept, *random* slope

Model	*df*	Deviance	vs.	Δ*df*	Δdeviance	*p*
M_0	3	6327.5				
M_{1a}	4	5823.8	M_0	1	503.69	< .001
$\boldsymbol{M_2}$	6	5770.7	M_{1a}	2	53.09	< .001
M_{1b}	4	6001.8	M_2	2	231.14	< .001

Adding Level-1 Predictors

Adding Level-2 Predictors

Example: popular2 (Hox, 2010, Appendix A)

- 2,000 students in 100 classrooms. Average class size is 20.
  - Popularity (*popular*: range = 0–10): average of popularity ratings by classmates.
  - Extraversion (*extrav*: range = 0–10)
  - Teacher experience (*texp*: years)

Adding Level-2 Predictors

Adding 'texp'

$$y_{ij} = \beta_{0j} + \beta_{1j} x_{ij} + e_{ij}$$
$$\beta_{0j} = \gamma_{00} + \gamma_{01} w_j + u_{0j}$$
$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$
$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{00} & \\ \tau_{10} & \tau_{11} \end{bmatrix}\right)$$

$$y_{ij} = \gamma_{00} + \gamma_{10} x_{ij} + \gamma_{01} w_j + u_{0j} + u_{1j} x_{ij} + e_{ij}$$

Adding Level-2 Predictors

Adding 'texp'

$$y_{ij} = \beta_{0j} + \beta_{1j} x_{ij} + e_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01} w_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{00} & \\ \tau_{10} & \tau_{11} \end{bmatrix}\right)$$

R:

```
> l1_rs_l2 ← lmer(popular ~ 1 + extrav + texp + (1 +
extrav|class), data=dat)
> summary(l1_rs_l2)
> confint(l1_rs_r2, method="profile", level=0.95)
```

- The terms before the parentheses describe the *fixed* components (parameters).
  - The "texp" represents the ***slope*** (γ_{01}) for teacher experience.

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: popular ~ 1 + extrav + texp + (1 + extrav | class)
   Data: dat

     AIC      BIC   logLik deviance df.resid
  5747.0   5786.2  -2866.5   5733.0     1993

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.0558 -0.7303  0.0116  0.6770  3.2064

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 class    (Intercept) 1.53006  1.2370
          extrav      0.03462  0.1861   -0.87
 Residual             0.89055  0.9437
Number of obs: 2000, groups:  class, 100

Fixed effects:
            Estimate Std. Error t value
(Intercept)  1.01780    0.22912   4.442
extrav       0.50226    0.02749  18.268
texp         0.09769    0.01016   9.617

Correlation of Fixed Effects:
       (Intr) extrav
extrav -0.721
texp   -0.701  0.104
> ### Confidence intervals of model parameters
> confint(l1_rs_l2, method="profile", level=0.95)
Computing profile confidence intervals ...
                  2.5 %     97.5 %
.sig01       0.93329826  1.5794570
.sig02      -0.93028015 -0.7401042
.sig03       0.13175439  0.2446508
.sigma       0.91377693  0.9752509
(Intercept)  0.50462689  1.5506855
extrav       0.44769468  0.5566324
texp         0.06980318  0.1251256
```

redictors

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + \gamma_{01} texp_j + u_{0j} + u_{1j} extrav_{ij} + e_{ij}$$

$\hat{\gamma}_{00} = 1.02\ (0.23),\ p < .001$

- The average popularity score for students whose extraversion score = 0 in a class of teachers who began teaching.

$\hat{\gamma}_{10} = 0.50\ (0.03),\ p < .001$

- The popularity score is expected to increase by 0.50 points with each 1-point increase in students' extraversion score *when controlling for teachers' experience*.

$\hat{\gamma}_{01} = 0.10\ (0.01),\ p < .001$

- The popularity score will increase by 0.10 points with each 1-year increase in teachers' experience *when controlling for students' extraversion score*.

Adding Level-2 Predictors

Adding 'texp'

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + \gamma_{01} texp_j + u_{0j} + u_{1j} extrav_{ij} + e_{ij}$$

```
Models:
l1_rs: popular ~ 1 + extrav + (1 + extrav | class)
l1_rs_l2: popular ~ 1 + extrav + texp + (1 + extrav | class)
         npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
l1_rs       6 5782.7 5816.3 -2885.3   5770.7
l1_rs_l2    7 5747.0 5786.2 -2866.5   5733.0 37.658  1  8.429e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- LRT $\chi^2(1) = 37.66$, $p < .001$ suggests that the current model fits significantly better than the previous model.
- This provides additional support for the effect of teacher experience on popularity.

Adding Level-2 Predictors

Adding 'extrav' x 'texp': Cross-level interaction

$$y_{ij} = \beta_{0j} + \beta_{1j} x_{ij} + e_{ij}$$
$$\beta_{0j} = \gamma_{00} + \gamma_{01} w_j + u_{0j}$$
$$\beta_{1j} = \gamma_{10} + \gamma_{11} w_j + u_{1j}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$
$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{00} & \\ \tau_{10} & \tau_{11} \end{bmatrix} \right)$$

$$y_{ij} = \gamma_{00} + \gamma_{10} x_{ij} + \gamma_{01} w_j + \gamma_{11} x_{ij} w_j + u_{0j} + u_{1j} x_{ij} + e_{ij}$$

*"main effect" of **x***

*"main effect" of **w***

*interaction of **x** and **w***

Adding Level-2 Predictors

Adding 'extrav' x 'texp': Cross-level interaction

$$y_{ij} = \beta_{0j} + \beta_{1j} x_{ij} + e_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01} w_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11} w_j + u_{1j}$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{00} & \\ \tau_{10} & \tau_{11} \end{bmatrix}\right)$$

R:

```
> l1_rs_l2_int ← lmer(popular ~ 1 + extrav + texp +
extrav*texp + (1 + extrav|class), data=dat)
> summary(l1_rs_l2_int)
> confint(l1_rs_r2_int, method="profile", level=0.95)
```

- The terms before the parentheses describe the *fixed* components (parameters).
  - The "extrav*texp" represents the ***slope*** (γ_{11}) for the interaction between student extraversion and teacher experience.

el-2 Predictors

```
Linear mixed model fit by maximum likelihood  ['lmerMo
Formula: popular ~ 1 + extrav + texp + extrav * texp +
   Data: dat

     AIC      BIC   logLik deviance df.resid
  5683.9   5728.7  -2834.0   5667.9     1992

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.2526 -0.7268  0.0161  0.6868  3.1898

Random effects:
 Groups   Name        Variance  Std.Dev. Corr
 class    (Intercept) 0.5179431 0.71968
          extrav      0.0003318 0.01822  -1.00
 Residual             0.8889094 0.94282
Number of obs: 2000, groups:  class, 100

Fixed effects:
             Estimate Std. Error t value
(Intercept) -1.211228   0.319743  -3.788
extrav       0.891135   0.044889  19.852
texp         0.251938   0.019664  12.812
extrav:texp -0.027465   0.002868  -9.576

Correlation of Fixed Effects:
            (Intr) extrav texp
extrav      -0.875
texp        -0.918  0.807
extrav:texp  0.774 -0.898 -0.861
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

> ### Confidence intervals of model parameters
> confint(l1_rs_l2_int, method="profile", level=0.95)
Computing profile confidence intervals ...
Warning: non-monotonic profile for .sig02Warning: bad
2.5 %      97.5 %
.sig01       0.49724455  0.97789573
.sig02      -1.00000000  1.00000000
.sig03       0.00000000  0.02576709
.sigma       0.91361313  0.97361458
(Intercept) -1.83975400 -0.57950438
extrav       0.80302892  0.97918124
texp         0.21299762  0.29058170
extrav:texp -0.03309265 -0.02182997
```

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + \gamma_{01} texp_j + \gamma_{11} extrav_{ij} texp_j + u_{0j} + u_{1j} extrav_{ij} + e_{ij}$$

$$\hat{\gamma}_{10} = 0.89\ (0.04),\ p < .001$$

$$\hat{\gamma}_{01} = 0.25\ (0.02),\ p < .001$$

$$\hat{\gamma}_{11} = -0.03\ (0.00),\ p < .001$$

- Students' popularity is influenced by both their extraversion and their teachers' experience in teaching. However, such impact diminishes as students are much extravert, teachers have taught many years, or both.

- The slope variance for 'extrav' (τ_{11}) is very small (0.0003). Thus, if we remove the level-1 slope variance (as well as the intercept-slope covariance) from the model…

Adding Level-2 Predictors

Final model

$$popular_{ij} = \gamma_{00} + \gamma_{10} extrav_{ij} + \gamma_{01} texp_j + \gamma_{11} extrav_{ij} texp_j + u_{0j} + e_{ij}$$

$$\left| \begin{array}{l} e_{ij} \sim N\left(0, \sigma_e^2\right) \\ u_{0j} \sim N\left(0, \tau_{00}\right) \end{array} \right.$$

$\hat{\gamma}_{10} = 0.89\ (0.04),\ p < .001$

$\hat{\gamma}_{01} = 0.25\ (0.02),\ p < .001$

$\hat{\gamma}_{11} = -0.03\ (0.00),\ p < .001$

Adding Level-2 Predictors

Final model

```
Models:
l1_fs_l2_int: popular ~ 1 + extrav + texp + extrav * texp + (1 | class)
l1_rs_l2_int: popular ~ 1 + extrav + texp + extrav * texp + (1 + extrav | class)
              npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
l1_fs_l2_int     6 5680.8 5714.4 -2834.4   5668.8
l1_rs_l2_int     8 5683.9 5728.7 -2834.0   5667.9 0.8558  2     0.6519
```

- LRT result is not significant. This new model *without* level-1 slope variance fits the data worse than the old model *with* it but "not significantly" worse — i.e., this model fits *as good as* the previous model. Thus, retain this model for the sake of *model parsimony*.

- Also, this new model provides the *smallest* ***AIC*** and ***BIC*** values compared to other models.

QUESTIONS? HELP?

Jason Lee

jaehoon.lee@ttu.edu

Woonyoung Song

woonyoung.song@ttu.edu